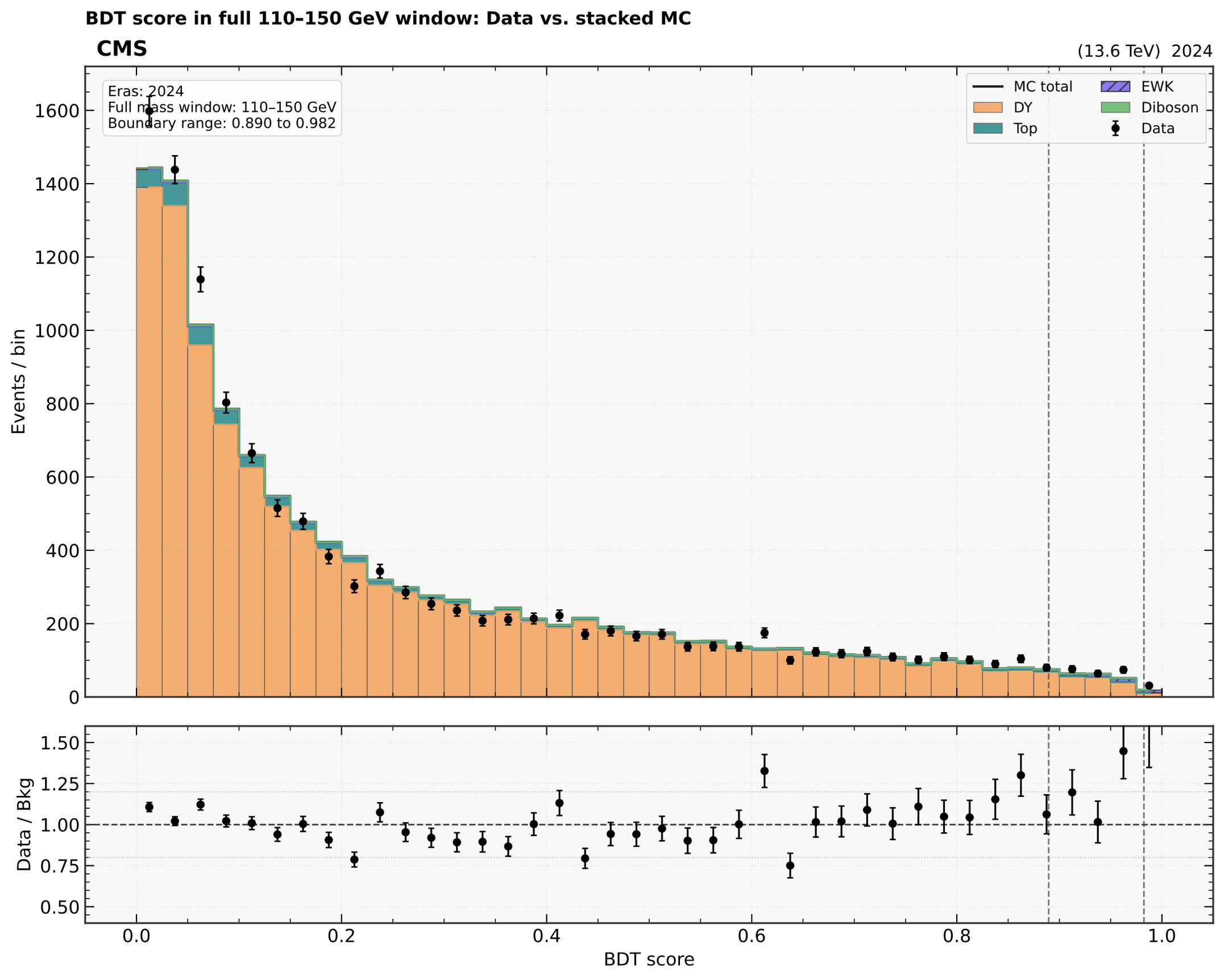
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 2024
Category edges : 0.0000, 0.8896, 0.9605, 0.9824, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

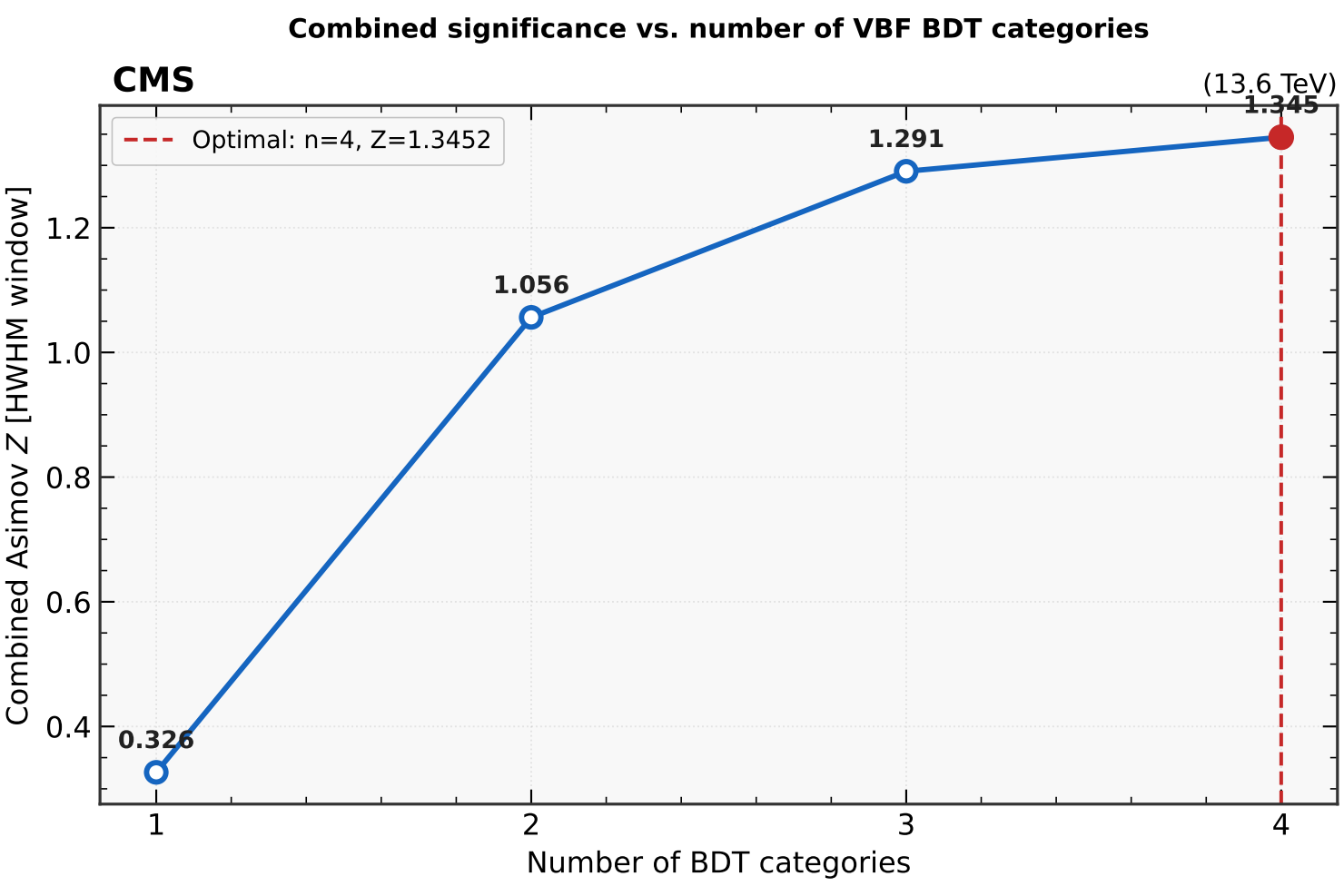

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 2024
Eras: 2024
Full mass window: 110–150 GeV
Boundary range: 0.890 to 0.982
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

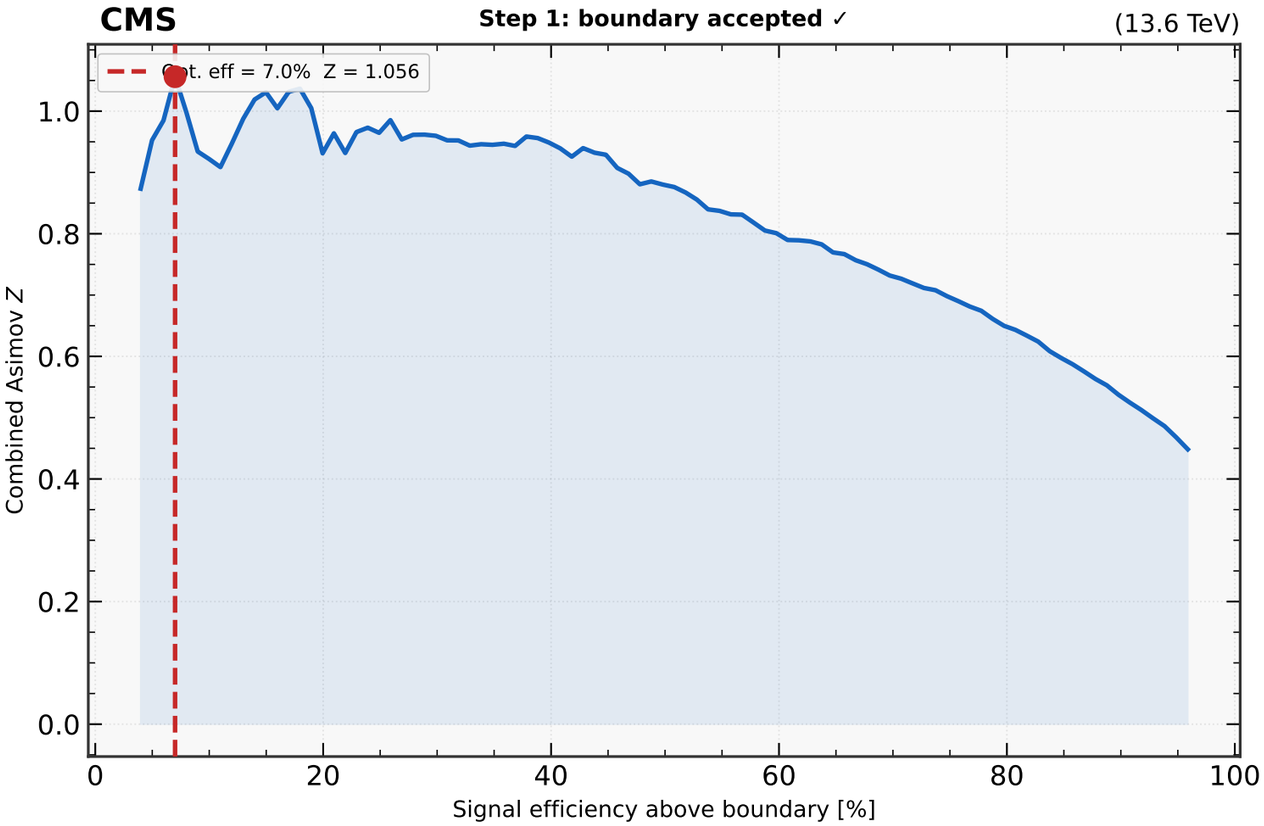
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.075	1443.48 ± 20.88	1598	0.01
[0.025, 0.050)	0.147	1407.92 ± 20.60	1438	0.01
[0.050, 0.075)	0.168	1015.07 ± 17.48	1139	0.02
[0.075, 0.100)	0.194	785.70 ± 15.34	803	0.02
[0.100, 0.125)	0.198	659.17 ± 14.14	665	0.03
[0.125, 0.150)	0.209	547.86 ± 12.87	515	0.04
[0.150, 0.175)	0.217	477.13 ± 12.25	479	0.05
[0.175, 0.200)	0.232	422.45 ± 11.53	383	0.05
[0.200, 0.225)	0.239	383.61 ± 11.29	302	0.06
[0.225, 0.250)	0.246	319.15 ± 9.79	343	0.08
[0.250, 0.275)	0.257	298.81 ± 9.52	285	0.09
[0.275, 0.300)	0.260	276.13 ± 9.23	254	0.09
[0.300, 0.325)	0.272	264.52 ± 9.04	236	0.10
[0.325, 0.350)	0.283	232.28 ± 8.29	208	0.12
[0.350, 0.375)	0.299	243.28 ± 8.71	211	0.12
[0.375, 0.400)	0.314	213.34 ± 8.05	214	0.15
[0.400, 0.425)	0.326	196.20 ± 7.72	222	0.17
[0.425, 0.450)	0.345	215.25 ± 8.64	171	0.16
[0.450, 0.475)	0.356	190.97 ± 7.64	180	0.19
[0.475, 0.500)	0.376	176.33 ± 7.39	166	0.21

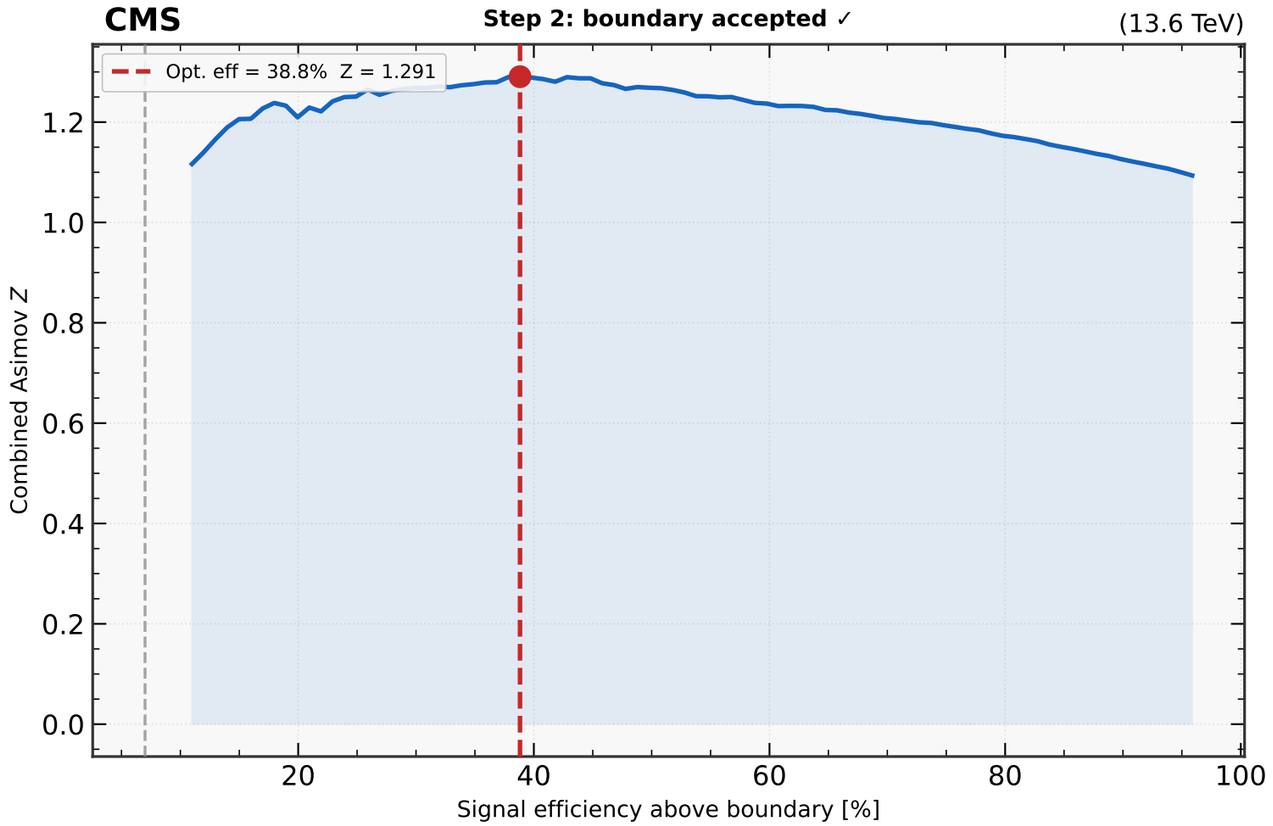
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.398	175.23 ± 7.37	171	0.23
[0.525, 0.550)	0.423	151.84 ± 6.69	137	0.28
[0.550, 0.575)	0.442	152.46 ± 6.90	138	0.29
[0.575, 0.600)	0.468	136.76 ± 6.39	137	0.34
[0.600, 0.625)	0.490	131.90 ± 6.25	175	0.37
[0.625, 0.650)	0.515	133.19 ± 6.22	100	0.39
[0.650, 0.675)	0.550	121.09 ± 5.94	123	0.45
[0.675, 0.700)	0.595	115.75 ± 5.59	118	0.51
[0.700, 0.725)	0.628	113.84 ± 5.81	124	0.55
[0.725, 0.750)	0.676	108.38 ± 5.58	109	0.62
[0.750, 0.775)	0.734	91.04 ± 4.93	101	0.80
[0.775, 0.800)	0.803	104.89 ± 5.63	110	0.76
[0.800, 0.825)	0.874	96.77 ± 5.41	101	0.90
[0.825, 0.850)	1.008	77.99 ± 4.65	90	1.28
[0.850, 0.875)	1.138	79.94 ± 4.76	104	1.40
[0.875, 0.900)	1.325	75.30 ± 4.63	80	1.73
[0.900, 0.925)	1.602	63.53 ± 3.99	76	2.46
[0.925, 0.950)	2.047	63.00 ± 4.08	64	3.15
[0.950, 0.975)	2.808	51.12 ± 3.67	74	5.21
[0.975, 1.000)	2.803	18.87 ± 1.76	31	12.93

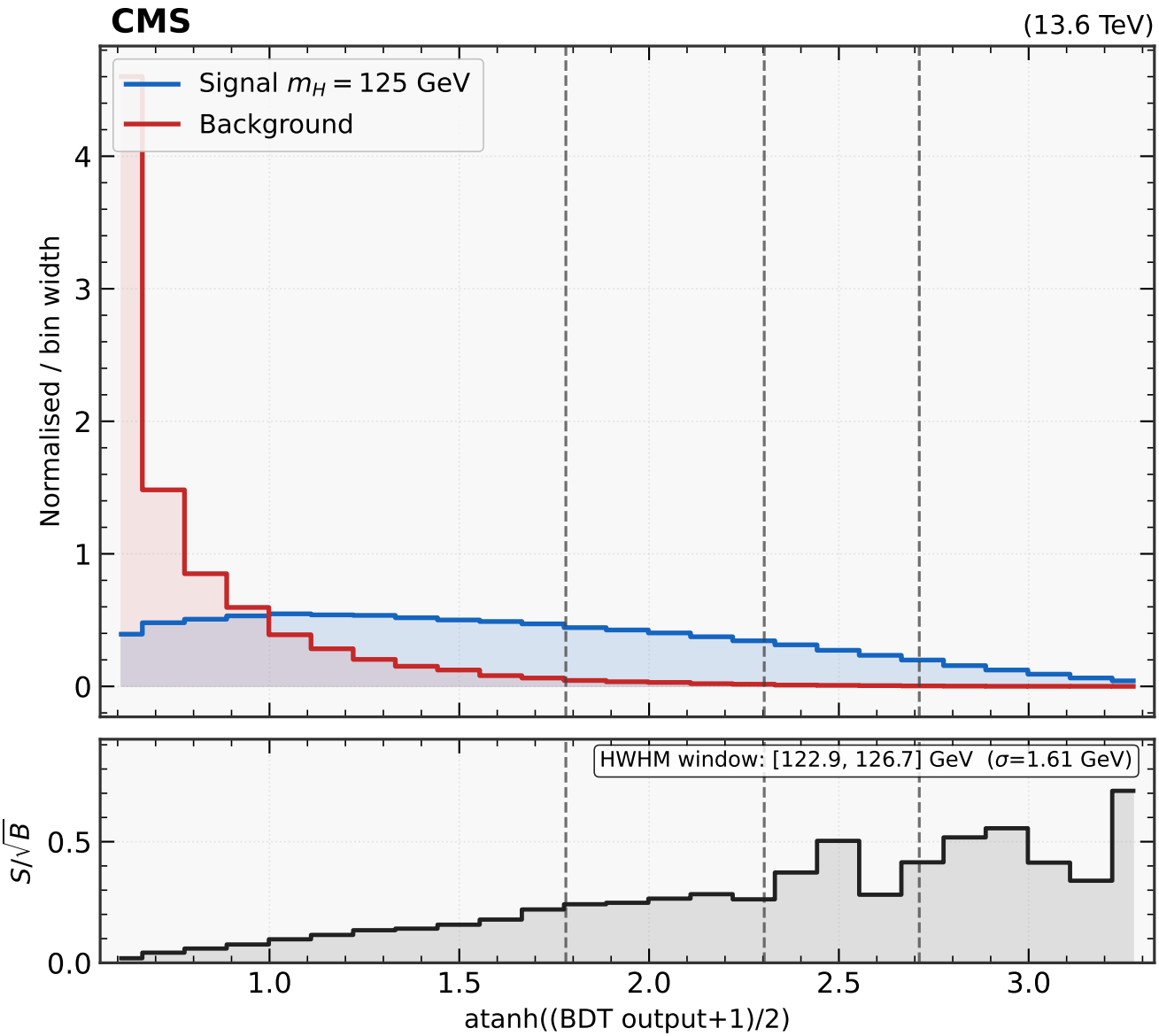

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=1.3452
0.326
1.056
1.291
1.345
Combined Asimov Z [HWHM window]
0.4
0.6
0.8
1.0
1.2
1
2
3
4
Number of BDT categories

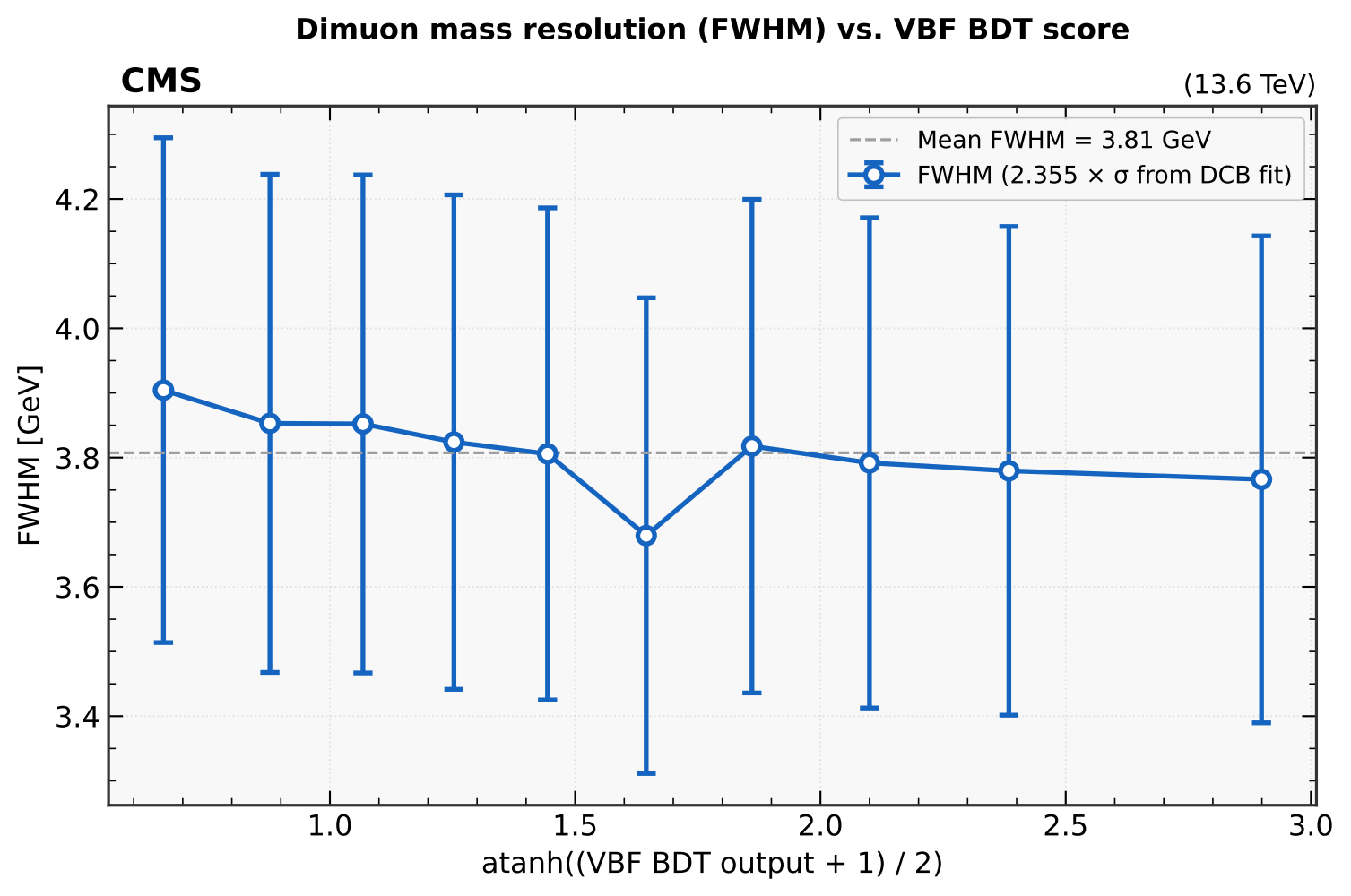
Iterative category boundary optimisation (AN Fig. 32 analog)

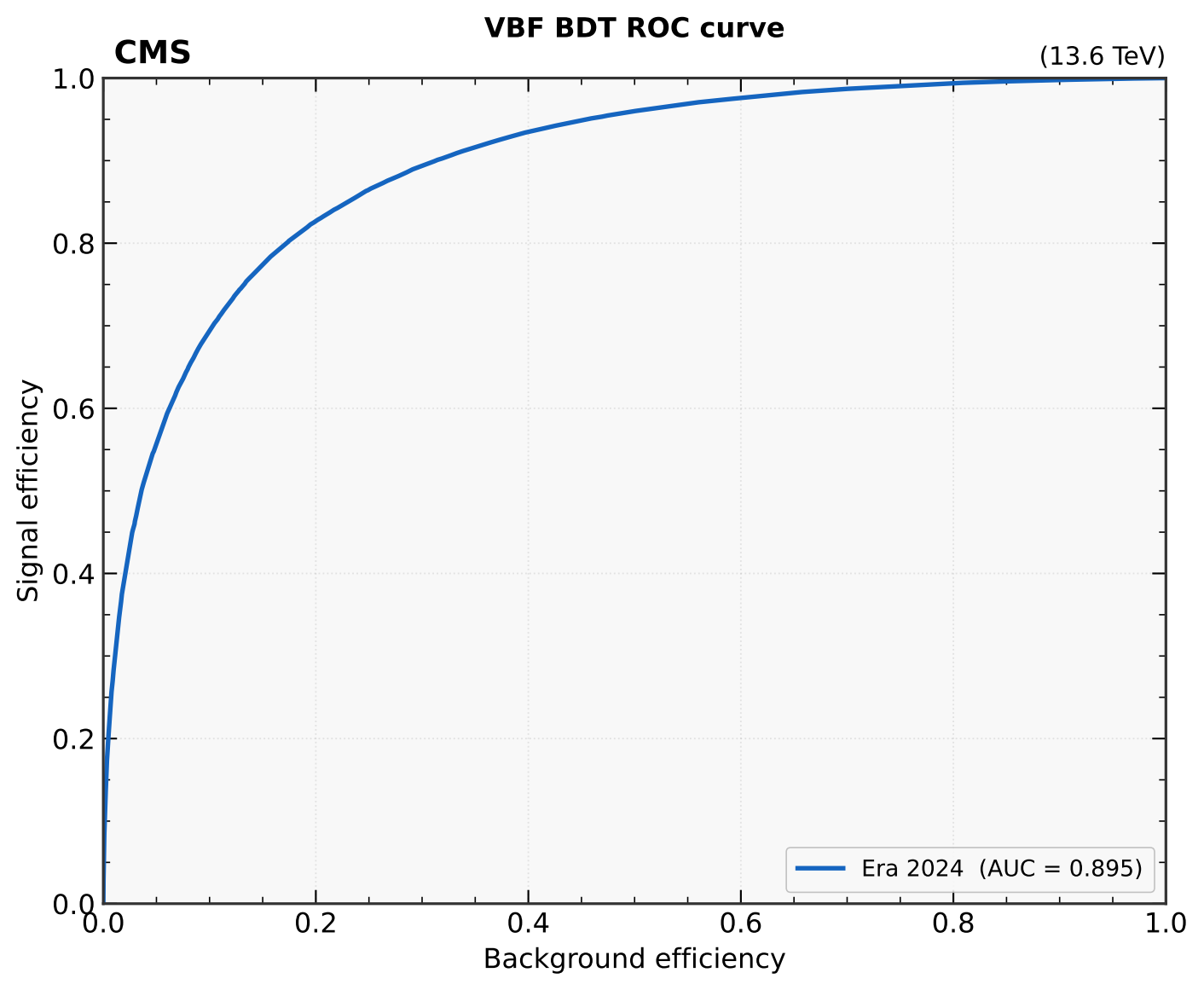


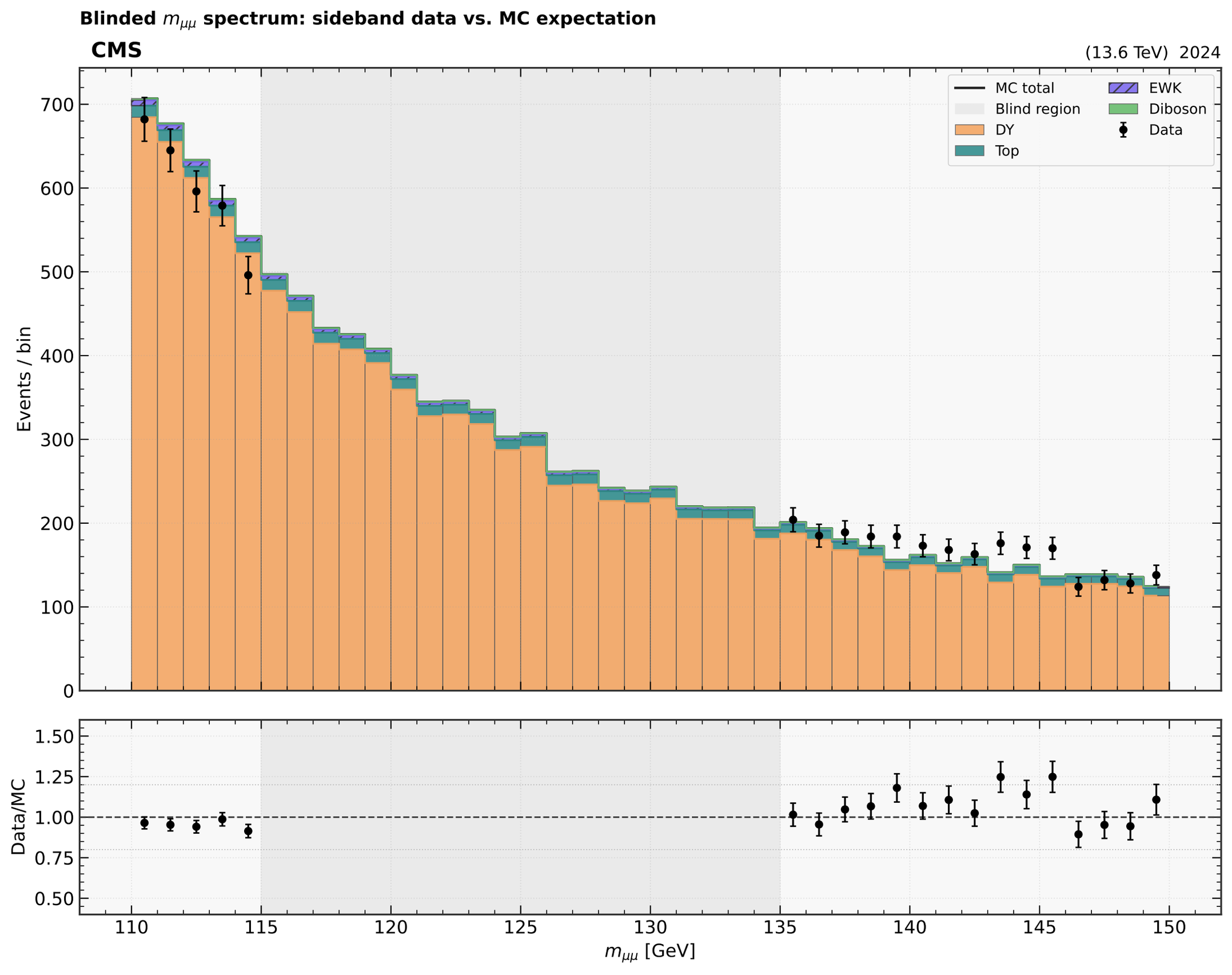


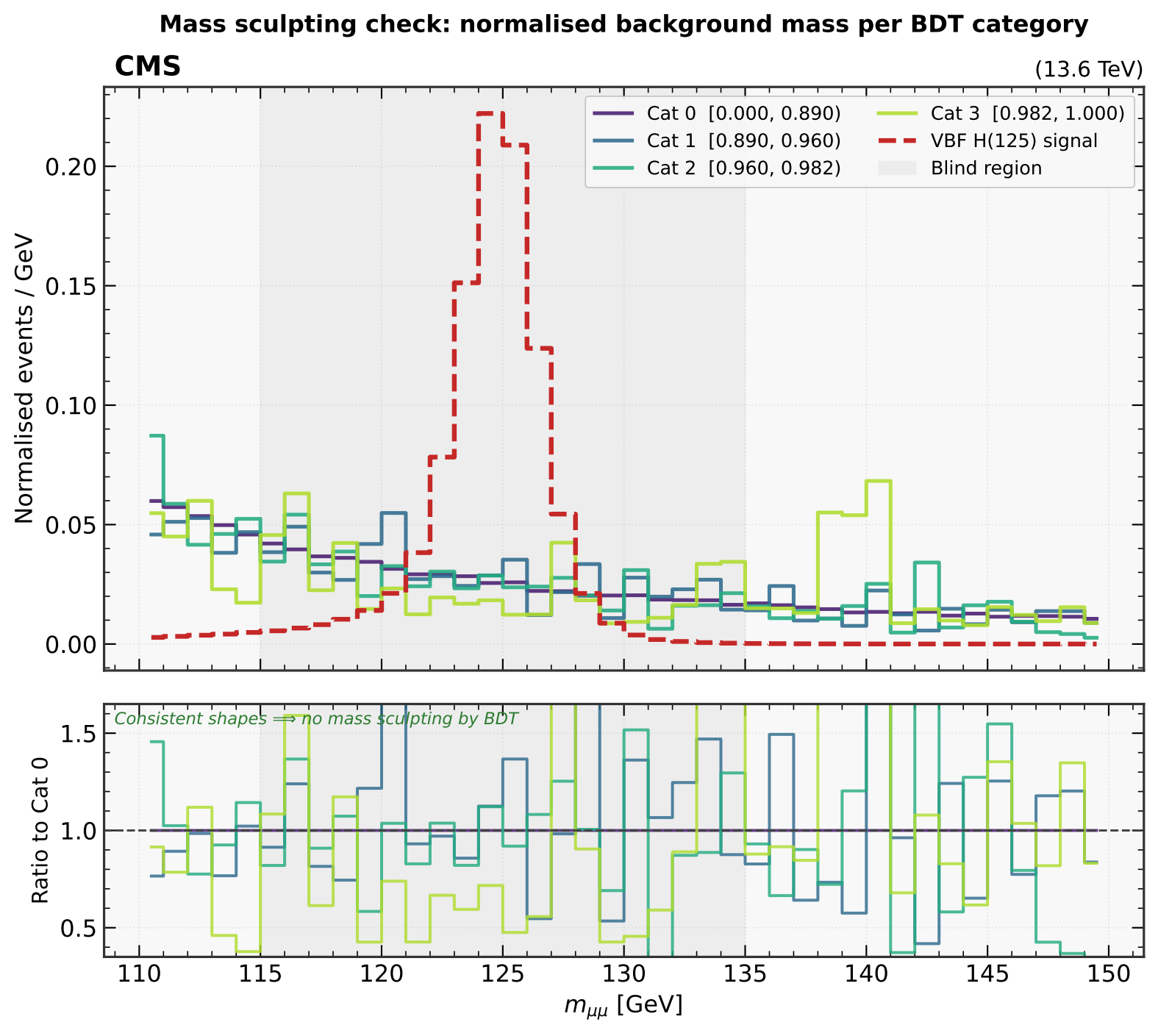
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window




Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
Mean FWHM = 3.81 GeV
FWHM (2.355 × σ from DCB fit)
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)

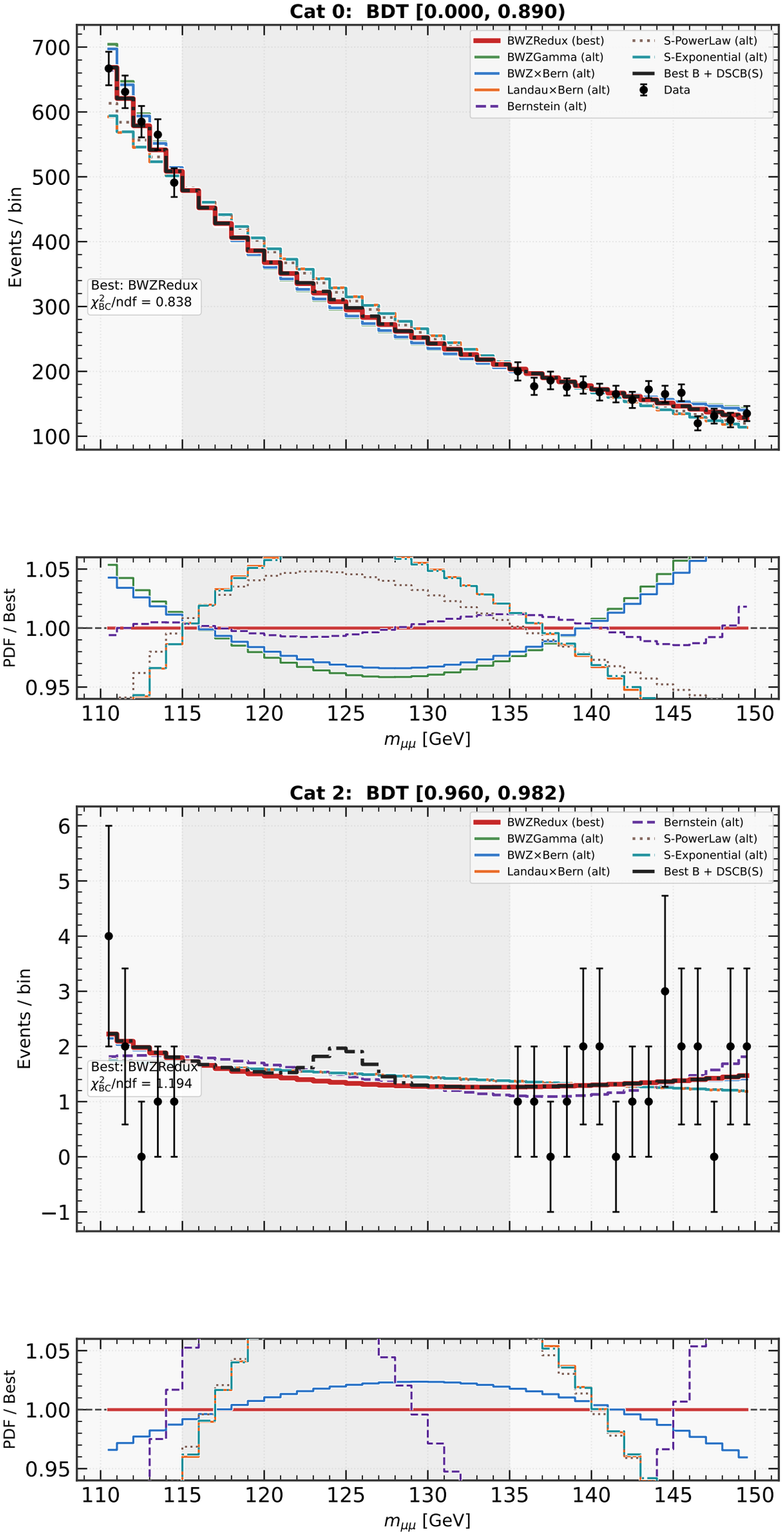

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 2024 (AUC = 0.895)


Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 2024
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

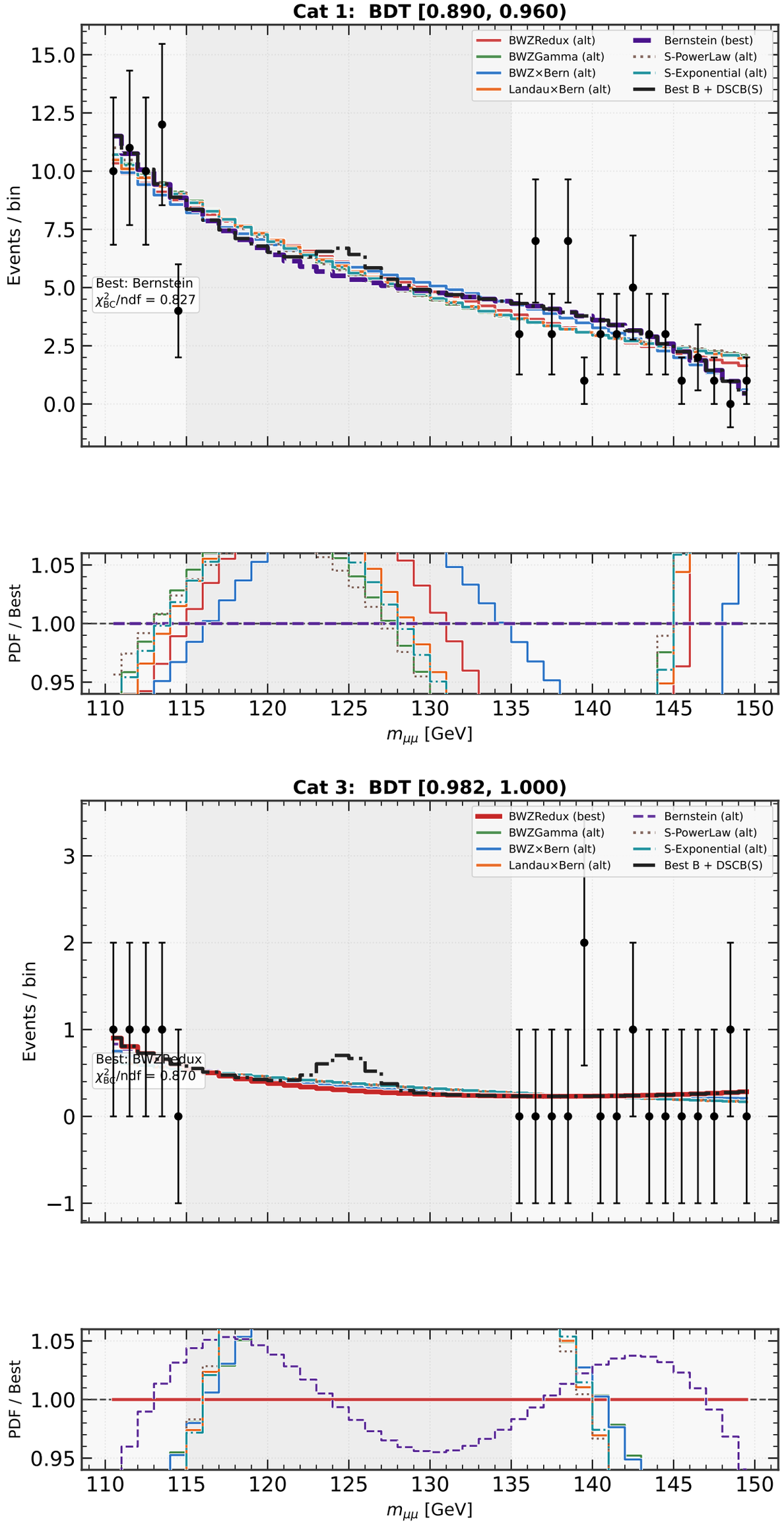

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.890)
Cat 1 [0.890, 0.960)
Cat 2 [0.960, 0.982)
Cat 3 [0.982, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⇒ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits

Cat 0: BDT [0.000, 0.890)



Cat 1: BDT [0.890, 0.960)



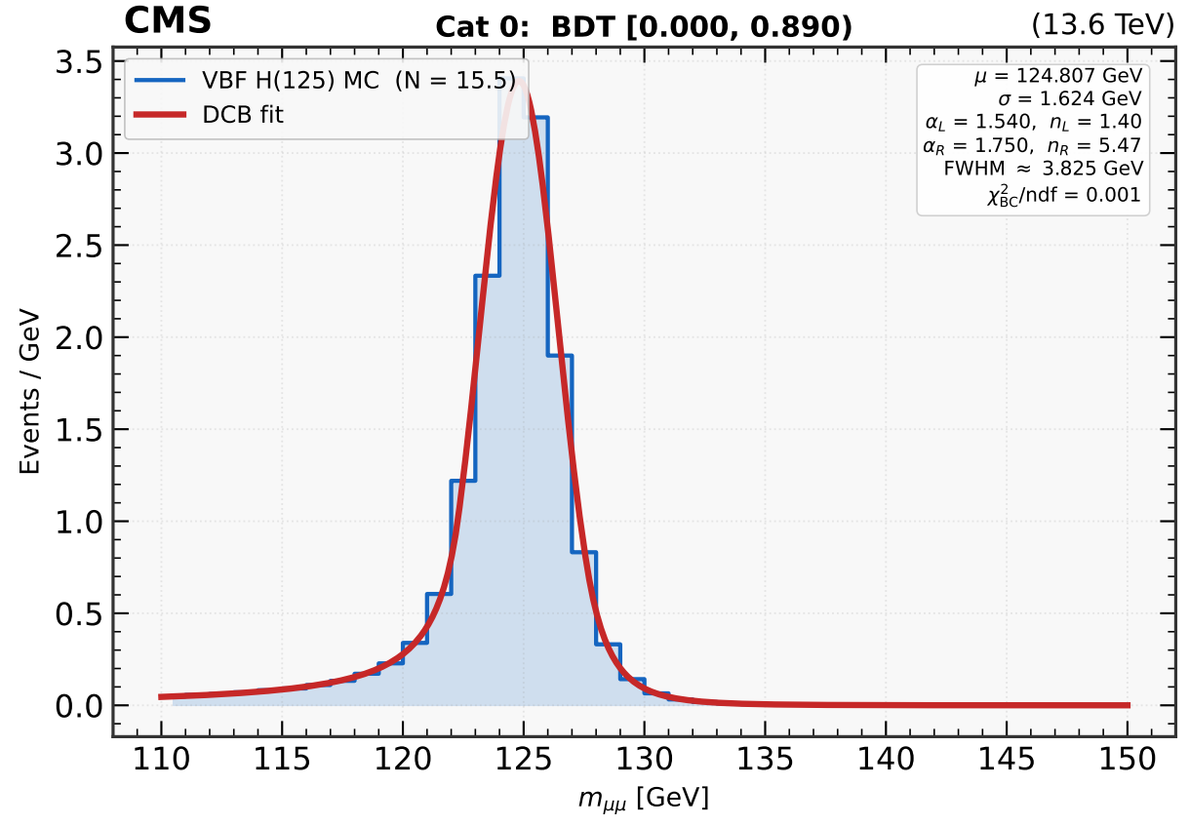
Cat 2: BDT [0.960, 0.982)

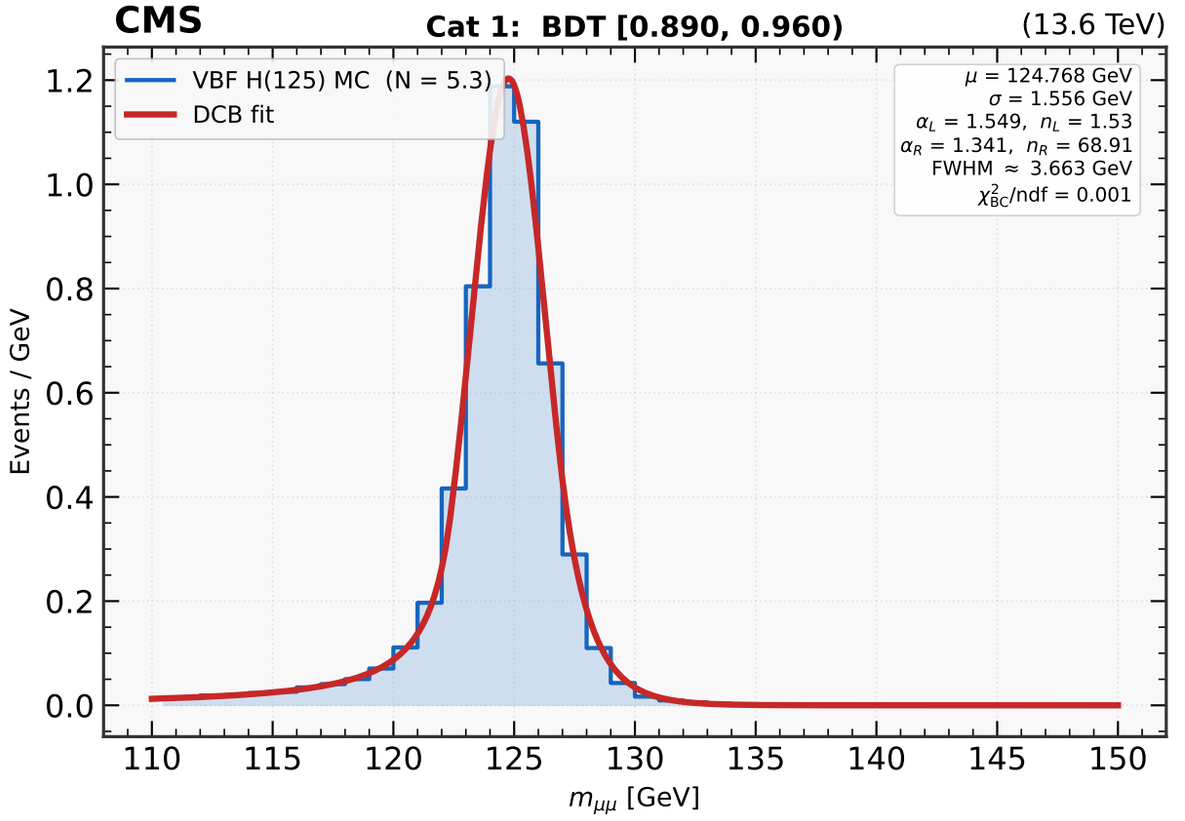
Cat 3: BDT [0.982, 1.000)

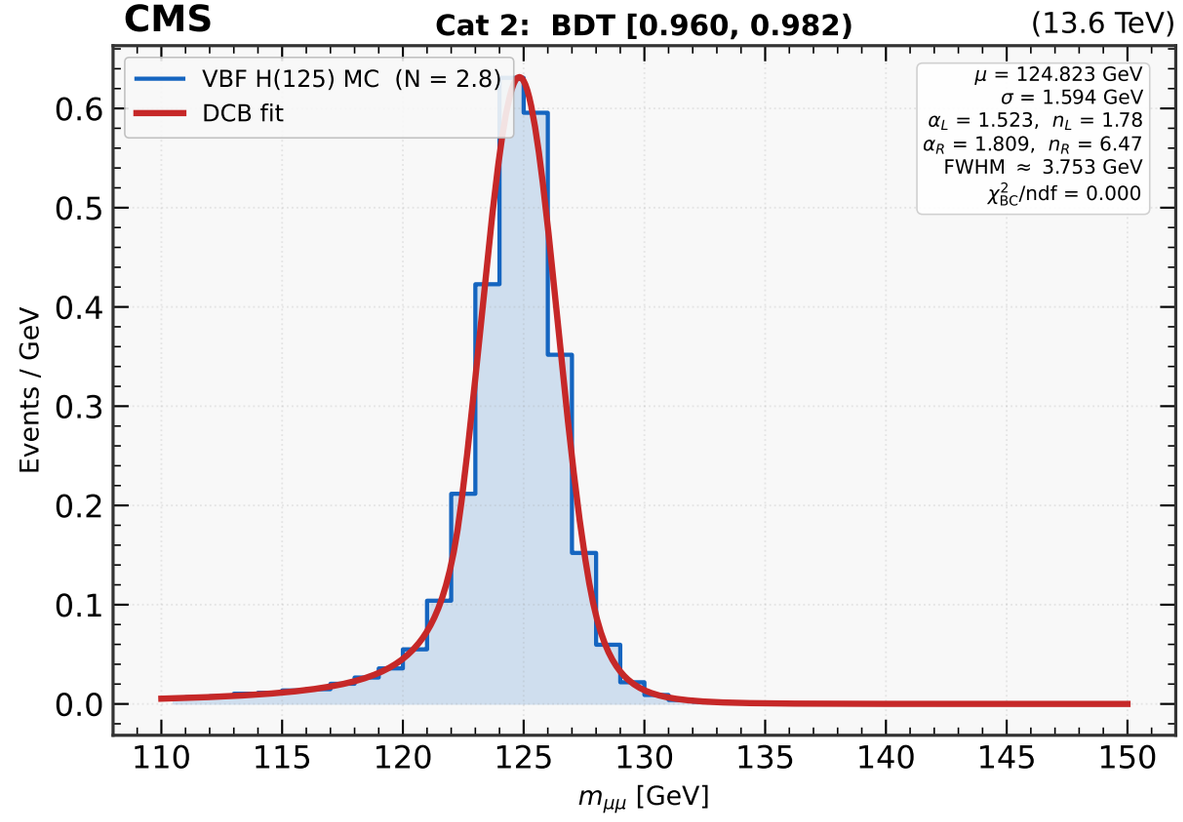
Per-category background model fit comparison (χ²_BC/ndf, ndf)

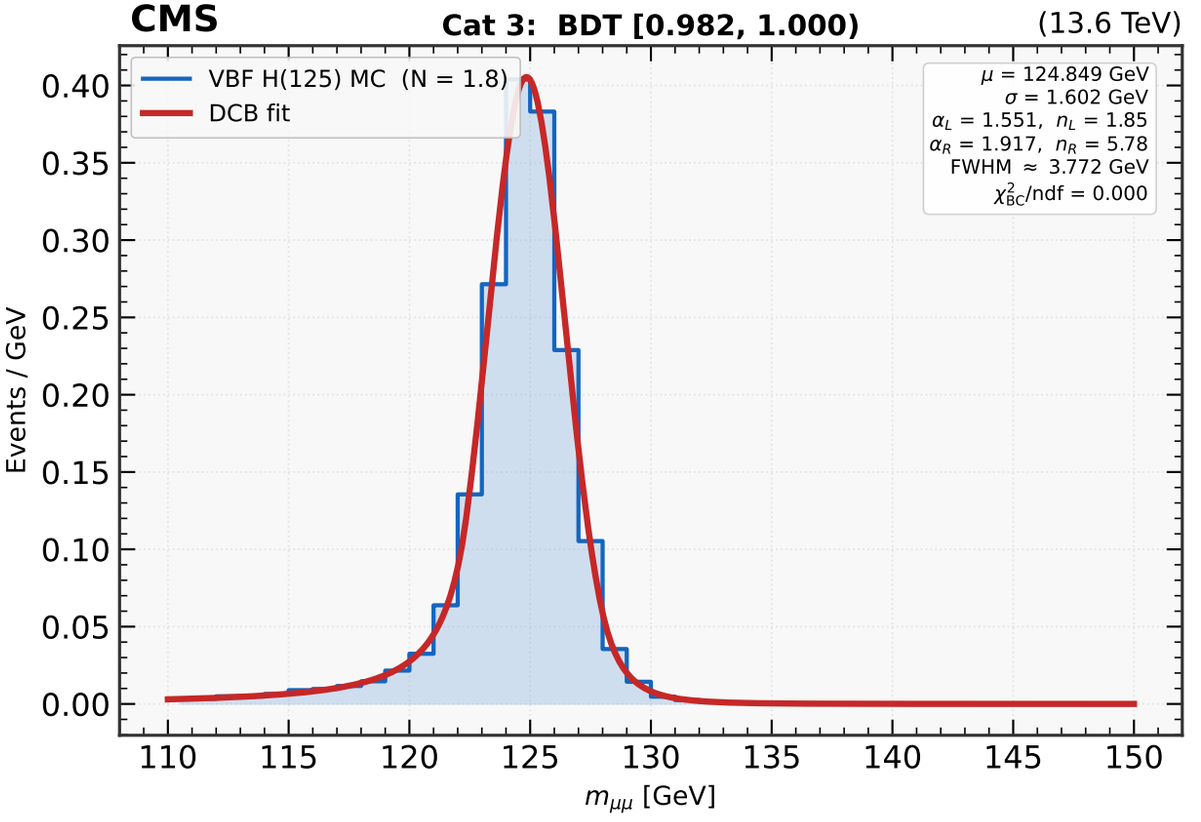
Category	BDT range	Best model	Best χ²_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.890)	BWZRedux	0.838	0.838 (37)	1.250 (37)	1.145 (36)	2.060 (37)	0.953 (34)	1.425 (37)	1.993 (37)
1	[0.890, 0.960)	Bernstein	0.827	0.906 (37)	0.949 (37)	0.860 (35)	0.938 (37)	0.827 (35)	0.949 (37)	0.939 (37)
2	[0.960, 0.982)	BWZRedux	1.194	1.194 (37)	1.195 (37)	1.228 (36)	1.207 (37)	1.205 (35)	1.204 (37)	1.205 (37)
3	[0.982, 1.000)	BWZRedux	0.870	0.870 (37)	0.876 (37)	0.901 (36)	0.890 (37)	0.920 (35)	0.887 (37)	0.890 (37)

VBF H(125) signal DCB extraction per BDT category



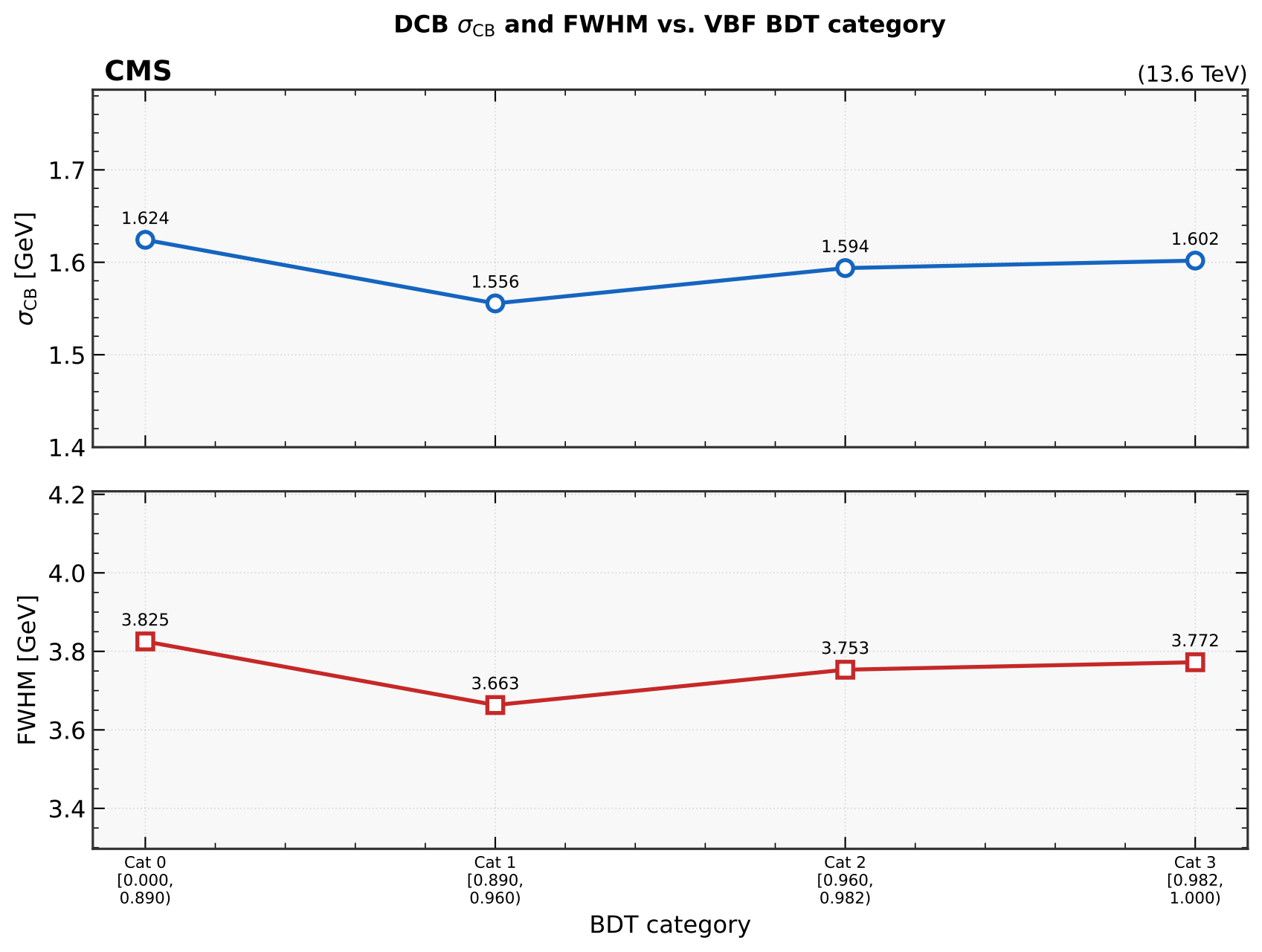


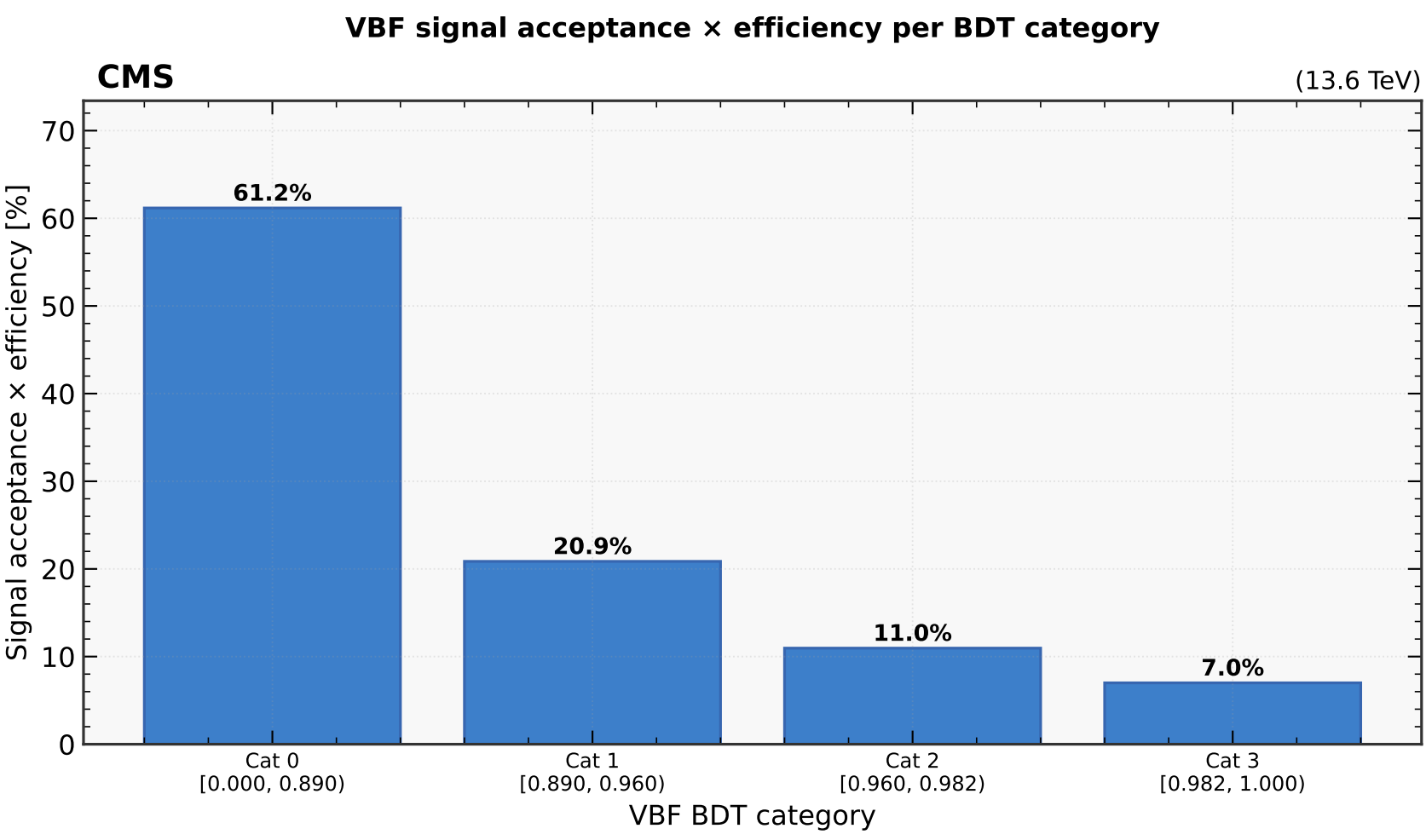




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ²_BC/ndf
0	[0.000, 0.890)	15.50	124.8067	1.6243	1.5401	1.395	1.7499	5.474	3.825	0.200	0.001
1	[0.890, 0.960)	5.29	124.7679	1.5555	1.5489	1.529	1.3405	68.905	3.663	0.545	0.001
2	[0.960, 0.982)	2.78	124.8229	1.5937	1.5234	1.778	1.8085	6.470	3.753	0.658	0.000
3	[0.982, 1.000)	1.77	124.8494	1.6018	1.5510	1.846	1.9169	5.780	3.772	1.023	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.7
1.6
1.5
1.4
1.624
1.556
1.594
1.602
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
3.825
3.663
3.753
3.772
Cat 0 [0.000, 0.890)
Cat 1 [0.890, 0.960)
Cat 2 [0.960, 0.982)
Cat 3 [0.982, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
10
20
30
40
50
60
70
61.2%
20.9%
11.0%
7.0%
Cat 0
[0.000, 0.890)
Cat 1
[0.890, 0.960)
Cat 2
[0.960, 0.982)
Cat 3
[0.982, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.890)	15.50	28414.65	10.609	2816.156	[122.9, 126.7]	0.200	0.200
1	[0.890, 0.960)	5.29	446.50	3.600	42.440	[122.9, 126.6]	0.553	0.545
2	[0.960, 0.982)	2.78	85.74	1.937	8.035	[122.9, 126.7]	0.683	0.658
3	[0.982, 1.000)	1.77	19.30	1.249	1.128	[123.0, 126.7]	1.176	1.023

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.890)	5361	13192.57 ± 98.51	0.406	28414.65 ± 145.36	0.1999	0.1998	2816.156
1	[0.890, 0.960)	90	189.66 ± 10.72	0.475	446.50 ± 17.22	0.5526	0.5450	42.440
2	[0.960, 0.982)	28	41.08 ± 5.10	0.682	85.74 ± 6.77	0.6833	0.6582	8.035
3	[0.982, 1.000)	8	10.09 ± 1.72	0.793	19.30 ± 2.11	1.1762	1.0227	1.128
Combined	all						1.3477	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.890)	27020.54	210.17	1089.53	94.41	28414.65	21130.19
1	[0.890, 0.960)	383.89	50.26	10.64	1.71	446.50	296.38
2	[0.960, 0.982)	59.44	23.87	2.14	0.30	85.74	45.87
3	[0.982, 1.000)	7.54	11.14	0.58	0.05	19.30	4.45